



Using lists to display outcomes for two events

Task A: Possible outcomes from a two stage trial

1) Set 1 Set 2

Jacob has listed some of the possible outcomes, when selecting one card from each set.

A☆	A♣	A*	D♥
C☆	B♦	B♠	B*
B♥	C♦	C♣	A♦
D*	B☆	D♣	C*
D♠	D♥	A♠	D♦

a) What outcomes has Jacob missed from his list?

$A♥, B♣, C♥, C♠, D☆$

b) What outcomes has Jacob duplicated?

$D♥$

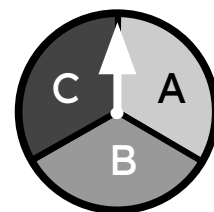
Task B: Systematic listing

1) A fair six-sided dice and the spinner shown are to be rolled and spun simultaneously.

a) List the possible outcomes **systematically**

$\{1A, 1B, 1C, 2A, 2B, 2C, 3A, 3B, 3C, 4A, 4B, 4C, 5A, 5B, 5C, 6A, 6B, 6C\}$

or $\{A1, A2, A3, A4, A5, A6, B1, B2, B3, B4, B5, B6, C1, C2, C3, C4, C5, C6\}$



b) How many possible outcomes are there?

18: 6 different outcomes from the dice and 3 different outcomes from the spinner

2) A fair coin and a fair six-sided dice are flipped and rolled simultaneously.

a) List the possible outcomes **systematically**

$\{1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T, 5H, 5T, 6H, 6T\}$

or $\{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$

b) How many possible outcomes are there?

12: 6 different outcomes from the dice and 2 different outcomes from the coin



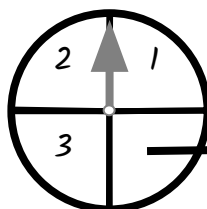
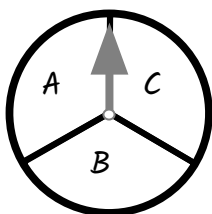
3) Two spinners are going to be spun.

The possible outcomes are

$\{A1, A2, A3, B1, B2, B3, C1, C2, C3\}$.

Complete the spinners.

e.g.



Either 1, 2 or 3

Task C: Listing outcomes of two events

1) Laura and Andeep are playing a game using this spinner. Laura wins a point if it lands on a **factor of 10** and Andeep wins a point if it lands on a **prime number**.

a) List the outcomes that win Laura a point.

$\{1, 2, 5\}$

b) List the outcomes that win Andeep a point.

$\{2, 3, 5\}$

c) List the outcomes for when neither Laura nor Andeep win a point.

$\{4\}$

d) List the outcomes for when Laura and Andeep both win a point.

$\{2, 5\}$

e) List the outcomes for when Laura wins a point, but Andeep does not.

$\{1\}$

